

Benchmarking Secure Multiparty Computation Protocols

Final talk for the Bachelor's Thesis by

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Chair of Network Architectures and Services
Department of Informatics
Technical University of Munich



Introduction

Use Case & Practicality of Secure Multiparty Computation

- General scenario [3]
 - Group of secret holders
 - Independent and private data
 - Distributed computation of a new piece of information
 - Concealing the secret inputs

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- Examples [1]
 - Economic: Determine and prevent an investment collision
 - Military: Compute target location
 - Research: Privacy preserving computation on secret data sets

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- Examples [1]
 - Economic: Determine and prevent an investment collision
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 - Research: Privacy preserving computation on secret data sets
- Requirements for Practicality of SMC
 - Low runtime
 - Effective host hardware (weakest node)
 - Adjusted network environment (weakest link)
 - Suitable protocol candidate

Research Questions

Aspects of SMC Performance

- What is the performance of state-of-the-art SMC protocols for different reference use cases?
- How does the performance differ when using protocols of different computation domains and adversary models?
- How are environmental parameters on the host, like processing capabilities, impacting the performance of different protocols?

Related Work

MP-SPDZ SMC Protocol Collection

SMC protocol source code collection:

MP-SPDZ [2] <https://github.com/data61/MP-SPDZ> by M. Keller being actively developed

Adversary model	Field ($\mathbb{F}_p$)	Ring ($\mathbb{Z}_{2^k}$)	Binary Secret sharing	Binary Garbled circuits
Malicious, dishonest majority				
Semi-honest, dishonest majority				
Malicious, honest majority				
Semi-honest, honest majority				

Table 1: SMC Adversary Models and Computation Domains

Related Work

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Table 2: Computation domains of *MP-SPDZ*' protocols

Sevarebench

Functionality

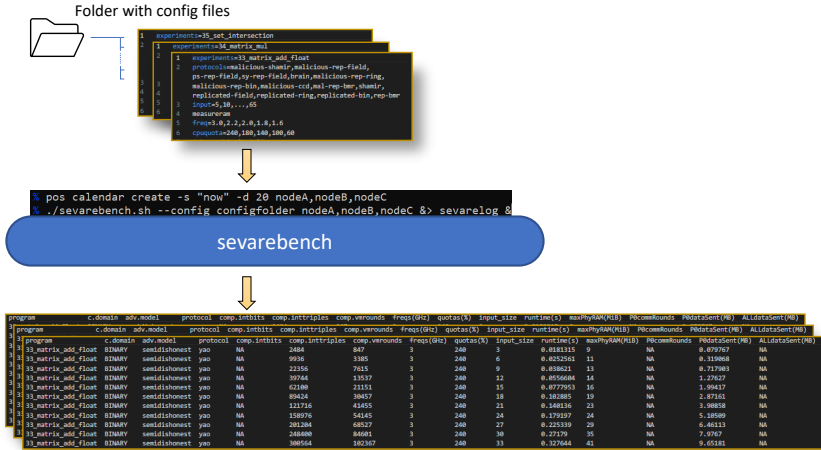


Figure 1: Benchmarking framework sevarebench usage

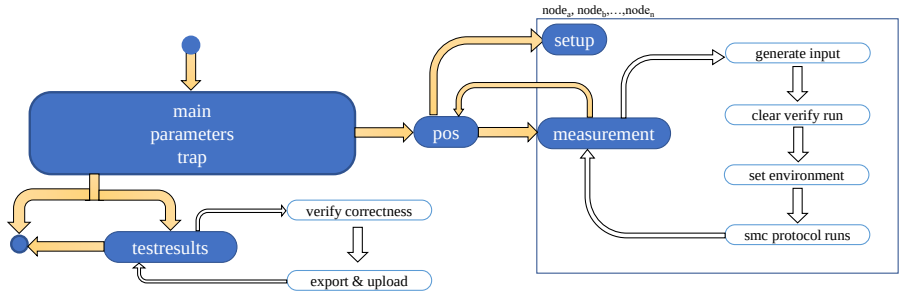


Figure 2: Modules of benchmarking framework sevarebench

Benchmarking SMC Protocols

"The Performance for Different Reference Use Cases"

"The performance of protocols from different computation domains"

- Choosing the Most Suitable Computation Domain
- Adversary Model: *Malicious, Honest Majority* Protocols
- Reference Problem: Fix-point addition

Benchmarking SMC Protocols

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Benchmarking SMC Protocols

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Adversary Model: *Malicious, Honest Majority* Protocols

Reference Problem: Fix-point addition

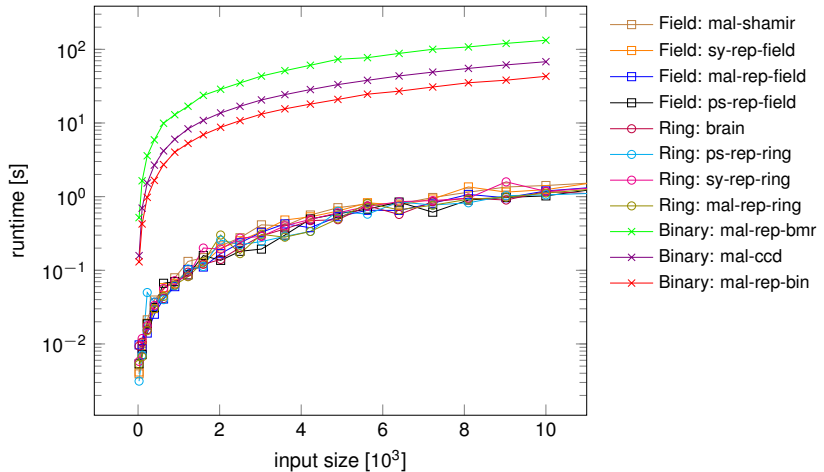


Figure 3: Fix-point addition - Compare Security Cost - Runtime

Benchmarking SMC Protocols

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Benchmarking SMC Protocols

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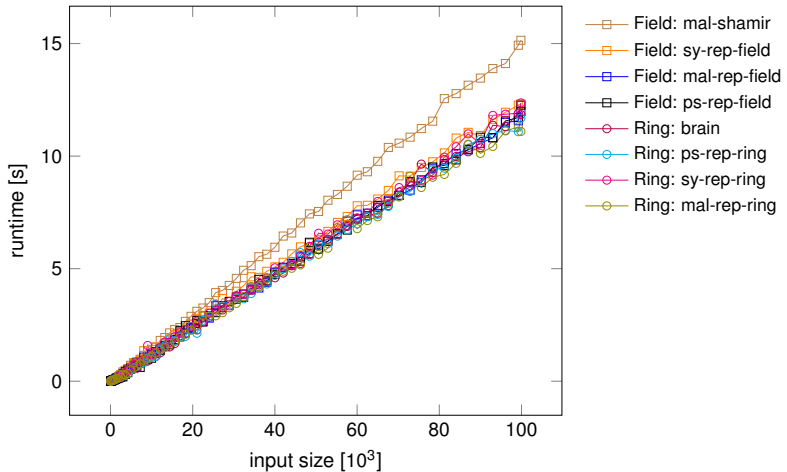


Figure 4: Fix-point addition - Compare Security Cost - Runtime

Benchmarking SMC Protocols

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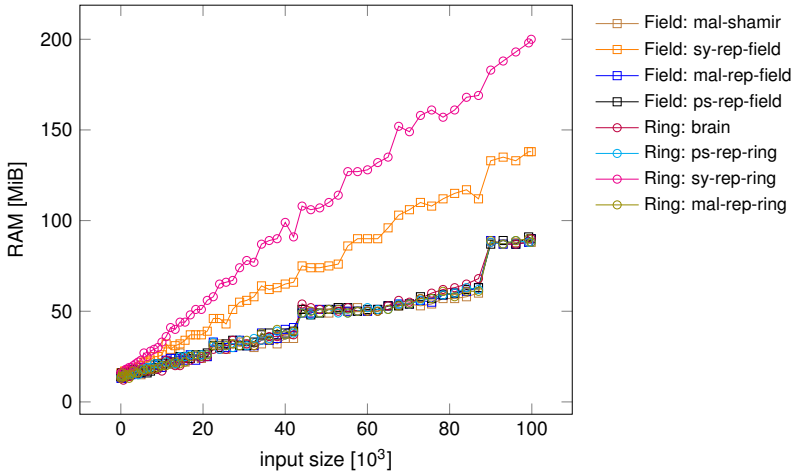


Figure 5: Fix-point addition - Compare Security Cost - RAM

Benchmarking SMC Protocols

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Table 5: Computation domains of *MP-SPDZ'* protocols - Most suitable *Malicious, Honest Majority* candidates for Fix-point Addition

Benchmarking SMC Protocols

"The Performance for Different Reference Use Cases"

"The performance of protocols from different adversary models"

- The Cost of Stronger Security
- Comparing the Different Adversary Models
- Reference Problem: Scalable Search

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Table 6: Computation domains of *MP-SPDZ'* protocols - Most suitable candidates for Scalable Search

Benchmarking SMC Protocols

The Cost of Stronger Security

RAM Measurements

Reference Problem: Scalable Search

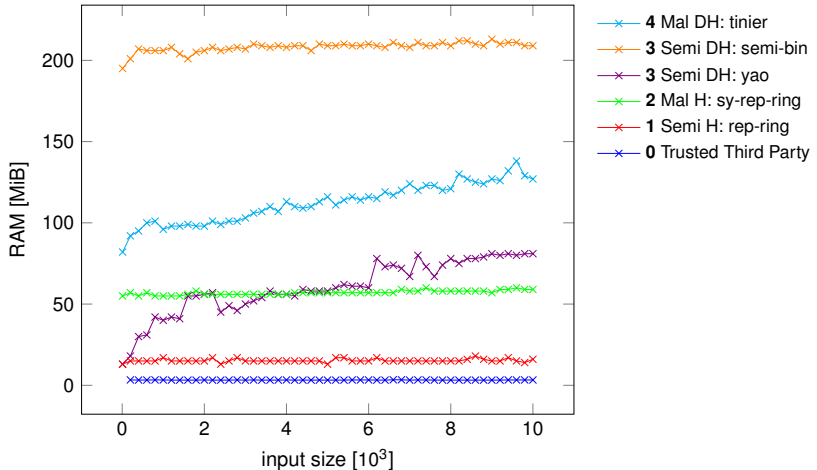


Figure 6: Scalable Search - Compare Security Cost - RAM

Benchmarking SMC Protocols

The Cost of Stronger Security

Runtime Measurements

Reference Problem: Scalable Search

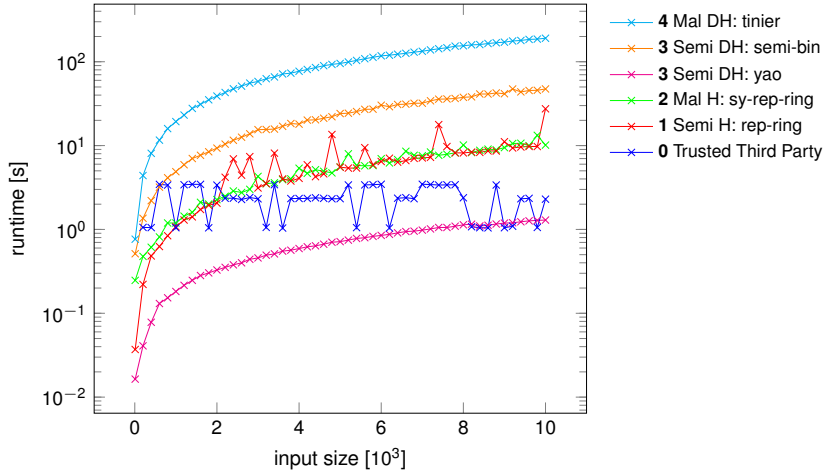


Figure 7: Scalable Search - Runtime Measurements

Benchmarking SMC Protocols

The Cost of Stronger Security

Runtime Interpolation

Reference Problem: Scalable Search

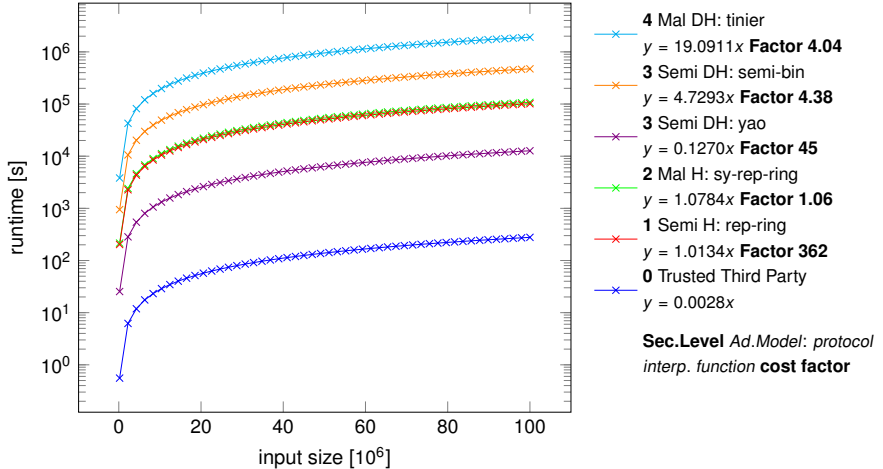


Figure 8: Scalable Search - Compare Security Cost - Runtime Interpolation

Benchmarking SMC Protocols

"The Performance for Different Reference Use Cases"

"CPU parameter impact on protocol performance"

- Finding Bottlenecks
- Influence of different CPU frequencies and quotas
- Reference Problem: Set Intersection

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Table 7: Computation domains of *MP-SPDZ*' protocols

Benchmarking SMC Protocols

CPU Frequency Influence¹

Adversary Model: *Honest Majority* SMC Protocols

Reference Problem: Set Intersection

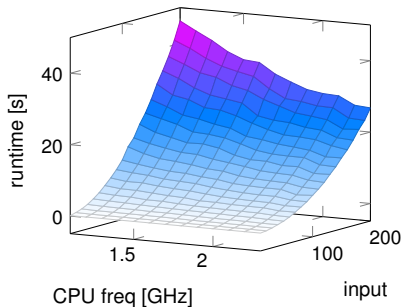


Figure 9: Replicated-bin on todd

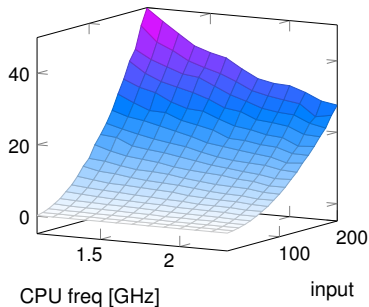


Figure 10: Malicious-rep-bin on todd

Interpolation at 1.1 GHz: $y = 0.2430x$

Interpolation at 2.2 GHz: $y = 0.1402x$

$2freq \rightarrow 0.57runtime$

Interpolation at 1.1 GHz: $y = 0.2599x$

Interpolation at 2.2 GHz: $y = 0.1457x$

$2freq \rightarrow 0.56runtime$

¹ Intel Xeon D-1518 2.2GHz

Benchmarking SMC Protocols

CPU Frequency Influence²

Adversary Model: *Semi-honest, Dishonest Majority* SMC Protocols

Reference Problem: Set Intersection

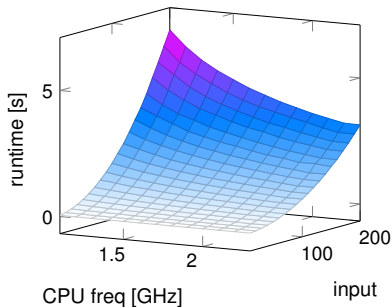


Figure 11: Yao on todd

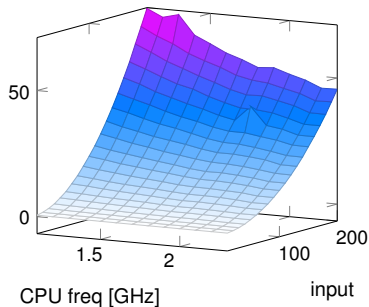


Figure 12: Semi-bin on todd

Interpolation at 1.1 GHz: $y = 0.0323x$

Interpolation at 2.2 GHz: $y = 0.0163x$

$2freq \rightarrow 0.51runtime$

Interpolation at 1.1 GHz: $y = 0.3704x$

Interpolation at 2.2 GHz: $y = 0.2372x$

$2freq \rightarrow 0.64runtime$

² Intel Xeon D-1518 2.2GHz

Benchmarking SMC Protocols

CPU Quota Influence³

Adversary Model: *Honest Majority* SMC Protocols

Reference Problem: Set Intersection

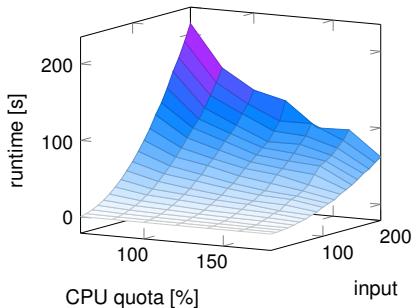


Figure 13: Replicated-bin on todd

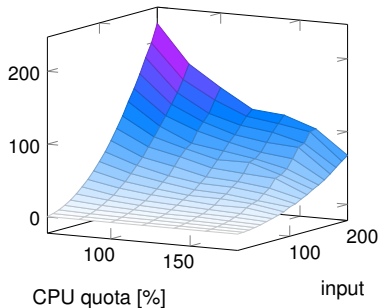


Figure 14: Malicious-rep-bin on todd

Interpolation at 60%: $y = 1.1242x$
 Interpolation at 120%: $y = 0.5978x$
 $2quota \rightarrow 0.53runtime$

Interpolation at 60%: $y = 1.1742x$
 Interpolation at 120%: $y = 0.6480x$
 $2quota \rightarrow 0.55runtime$

³ Intel Xeon D-1518 2.2GHz

Benchmarking SMC Protocols

CPU Quota Influence⁴

Adversary Model: *Semi-honest, Dishonest Majority* SMC Protocols

Reference Problem: Set Intersection

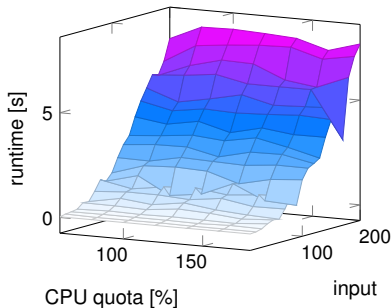


Figure 15: Yao on todd

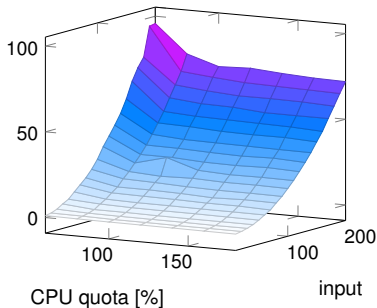


Figure 16: Semi-bin on todd

Interpolation at 60%: $y = 0.0388x$
 Interpolation at 120%: $y = 0.0414x$
 $2quota \rightarrow 1.07runtime$

Interpolation at 60%: $y = 0.5172x$
 Interpolation at 120%: $y = 0.3896x$
 $2quota \rightarrow 0.75runtime$

⁴ Intel Xeon D-1518 2.2GHz

Conclusion

Analyzing the performance of SMC protocols

Sevarebench

- Simplifies SMC protocol benchmarking
- + MP-SPDZ: Offers many state-of-the art protocol implementations
- + pos: Makes measurements reproducible
- Allows the comparison of many SMC categories
- Error handling, extendable, configurable

Research Answers

- Choosing the Most Suitable Computation Domain
 - Reference Problem: Fix-point addition → **Field, Ring**
 - Reference Problem: Set Intersection → **Binary**
- The Cost of Stronger Security
 - Reference Problem: Scalable Search
 - SMC Overhead: Factor **364** (Yao **45**)
 - Honest Majority → Dishonest Majority cost: Factor **4.38**
- CPU parameter impact - Finding Bottlenecks
 - Reference Problem: Set Intersection
 - Doubling frequency can reduce runtime between **36%** and **44%** (Yao **49%**)
 - Doubling quota can reduce runtime between **25%** and **47%** (Yao **no reduction**)

Conclusion

Related and Future Work

"Secure and Privacy-Preserving Services based on SMC" by Maltitz [4]

- SMC mainly communication bound
- Network latency strongest influence on runtime
- Number of participants influence
- CPU not notably challenged

→ Measurements show that CPU parameters influence performance having ideal networking conditions

"MP-SPDZ: A Versatile Framework for Multi-Party Computation" by Keller [2]

- Lower security requirements generally allow for more efficient protocols
- Field and Ring protocols offer integer addition and multiplication at low cost

→ Measurements confirm these findings

Future Work

- Compare optimized protocols
- Compare optimized SMC programs
- Include influence of network environment
- More complex reference use cases
- Benchmark bigger group sizes

Additional Data

Calculating the Cost of Stronger Security

- Determine the best performing protocol for each computation domain
- Run benchmark with same input sizes for these protocols
- Interpolate measurement results
- Compare function coefficients

Additional Data

Choosing the Most Suitable Computation Domain - Yao

Reference Problem: Fix-point addition

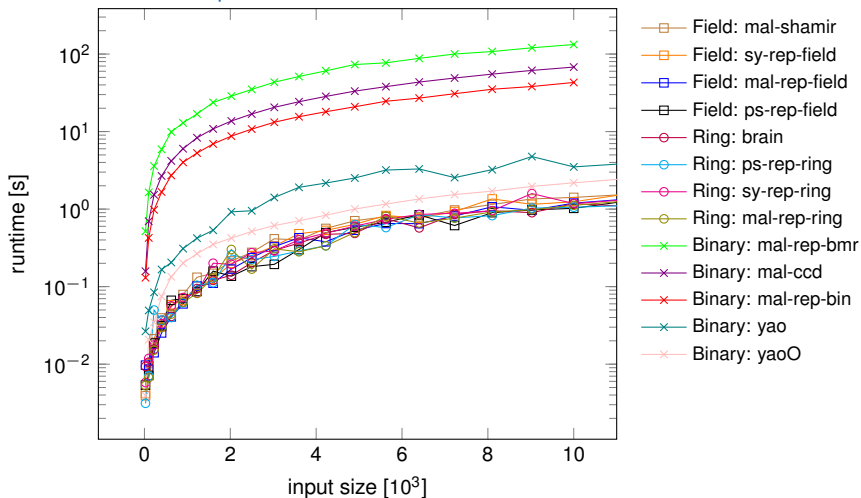


Figure 17: Fix-point addition - Compare Security Cost - Runtime

Additional Data

Choosing the Most Suitable Computation Domain 2 - Yao

Reference Problem: Fix-point addition

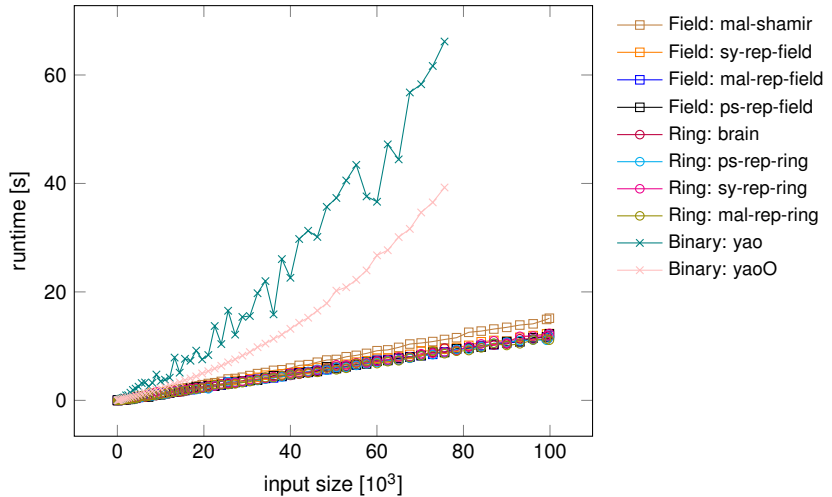


Figure 18: Fix-point addition - Compare Security Cost - Runtime

Additional Data

Choosing the Most Suitable Computation Domain 3 RAM - Yao

Reference Problem: Fix-point addition

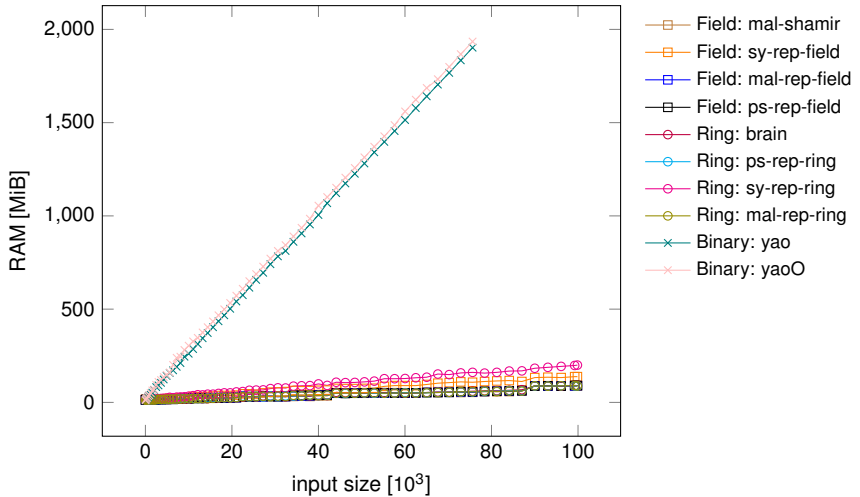


Figure 19: Fix-point addition - Compare Security Cost - RAM

Additional Data

The Cost of Stronger Security - Runtime - Yao

Reference Problem: Scalable Search

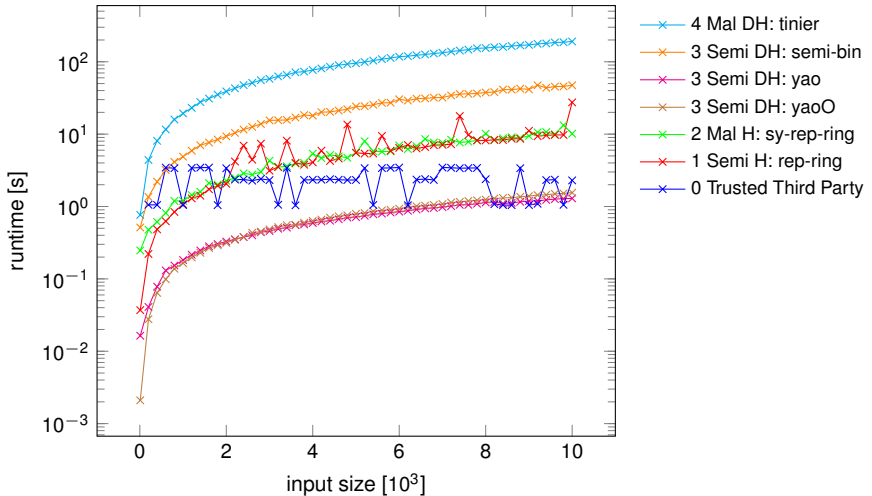


Figure 20: Scalable Search - Compare Security Cost - Runtime

Additional Data

The Cost of Stronger Security - Runtime - Yao

Reference Problem: Scalable Search

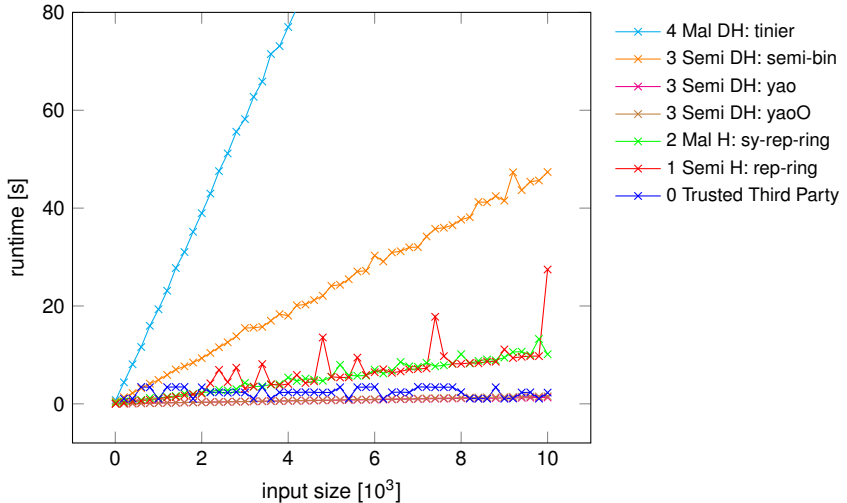


Figure 21: Scalable Search - Compare Security Cost - Runtime - Linear

Additional Data

TTP Unconcealed Computation in C++
Reference Problem: Scalable Search

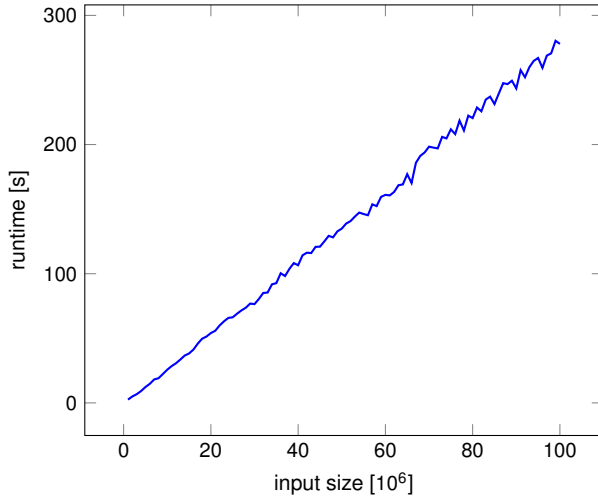


Figure 22: TTP computation of Scalable Search on todd

Additional Data

Benchmarking Framework Modules

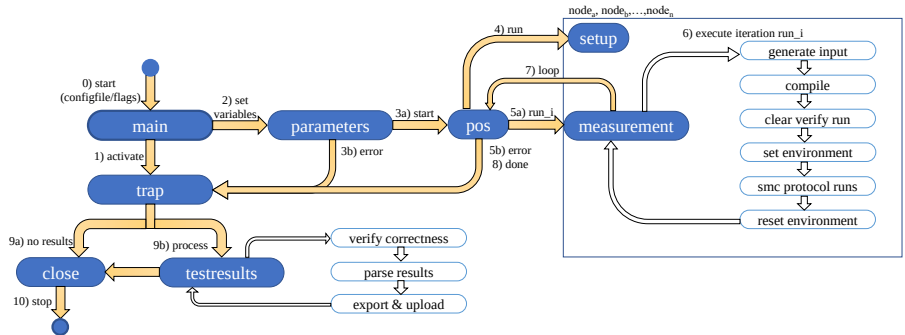


Figure 23: Modules of benchmarking framework sevarebench

- [1] W. Du and M. J. Atallah.
Secure multi-party computation problems and their applications: A review and open problems.
In Proceedings of the 2001 Workshop on New Security Paradigms, NSPW '01, page 13–22, New York, NY, USA, 2001.
Association for Computing Machinery.
- [2] M. Keller.
MP-SPDZ: A versatile framework for multi-party computation.
In Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security, 2020.
- [3] Y. Lindell.
Secure multiparty computation (mpc).
Cryptology ePrint Archive, Report 2020/300, 2020.
- [4] M. von Maltitz.
Secure and privacy-preserving services based on secure multiparty computation, 07 2019.